

Studio Shodwe



Instalasi Kubernetes



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Apa itu Kubernetes



Kubernetes adalah platform untuk mengelola container secara otomatis, seperti:

Menjalankan aplikasi container, Menjaga aplikasi tetap hidup,
Load balancing, Scaling otomatis, Self healing, Deploy multi server





Cara Install Kubernetes



Opsi 1 — Docker

Desktop Kubernetes

Cocok untuk user Windows yang ingin cepat mencoba.

Opsi 3 — Windows via Minikube (paling bagus untuk belajar di Windows)

Paling mudah dan fleksibel.

Opsi 2 — Linux via MicroK8s (paling bagus untuk Linux)

Ringan dan stabil.





Langkah I Buka PowerShell



Menjalankan Cluster

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> wsl
Windows Subsystem for Linux has no installed distributions.
You can resolve this by installing a distribution with the instructions below:

Use 'wsl.exe --list --online' to list available distributions
and 'wsl.exe --install <distro>' to install.
PS C:\WINDOWS\system32> wsl --install
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Create a default unix user account: jay
New password:
Retype new password:
password: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

jays@DESKTOP-STVHV12:~/kubernetes$ cd
jays@DESKTOP-STVHV12:~/kubernetes$ wsl
wsl: command not found
jays@DESKTOP-STVHV12:~/kubernetes$ exit
exit
PS C:\WINDOWS\system32> wsl . -v
NAME STATE VERSION
* Ubuntu Running 2
PS C:\WINDOWS\system32>
```

```
jays@DESKTOP-STVHV12:~$ minikube start --driver=docker
minikube v1.38.1 on Ubuntu 24.04 (kvm/amd64)
Using the docker driver based on user configuration
Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
Using Docker driver with root privileges
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.50 ...
Downloading Kubernetes v1.35.1 preload ...
> preloaded-images-k8s-v18-v1...: 272.45 MiB / 272.45 MiB 100.00% 5.45 Mi
> gcr.io/k8s-minikube/kicbase...: 519.58 MiB / 519.58 MiB 100.00% 6.57 Mi
Creating docker container (CPUs=2, Memory=3072MB) ...
Failing to connect to https://registry.k8s.io/ from inside the minikube container
To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
Configuring bridge CNI (Container Networking Interface) ...
Verifying Kubernetes components...
* Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: storage-provisioner, default-storageclass
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
jays@DESKTOP-STVHV12:~$
```

Cek node:

```
jays@DESKTOP-STVHV12:~/kubernetes$ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane	16m	v1.35.1
minikube-m02	NotReady	<none>	107s	v1.35.1



Struktur Project



Buat folder project :

```
jayy@DESKTOP-STVHVI2:~$ mkdir kubernetes  
jayy@DESKTOP-STVHVI2:~$ cd kubernetes  
jayy@DESKTOP-STVHVI2:~/kubernetes$ code .
```





Deployment & Service



Deployment

- Menjalankan container nginx
- Menentukan jumlah pod
- Auto restart jika crash

Service

- Menghubungkan user ke pod
- Load balancing



File Yaml

deployment.yaml

```
GNU nano 7.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
        - name: nginx
          image: nginx:alpine
          imagePullPolicy: IfNotPresent
          ports:
            - containerPort: 80
```

service.yaml

```
GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```



Menjalankan YAML



Apply File :

```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/web-nginx created
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl apply -f service.yaml
service/web-nginx-service created
```

Lihat Semua Resource :

```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/web-nginx-748d764f7-kf9cm	0/1	ContainerCreating	0	20s
pod/web-nginx-748d764f7-kwxtg	0/1	ErrImagePull	0	20s
pod/web-nginx-748d764f7-vftx5	0/1	ContainerCreating	0	20s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	6m5s
service/web-nginx-service	NodePort	10.105.31.87	<none>	80:30486/TCP	6s

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/web-nginx	0/3	3	0	20s

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/web-nginx-748d764f7	3	3	0	20s

Cek Pod dan Monitoring Live :

```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
web-nginx-748d764f7-kf9cm	0/1	ContainerCreating	0	31s
web-nginx-748d764f7-kwxtg	0/1	ImagePullBackOff	0	31s
web-nginx-748d764f7-vftx5	0/1	ErrImagePull	0	31s

```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl get pods -w
```

NAME	READY	STATUS	RESTARTS	AGE
web-nginx-748d764f7-kf9cm	0/1	ErrImagePull	0	44s
web-nginx-748d764f7-kwxtg	0/1	ImagePullBackOff	0	44s
web-nginx-748d764f7-vftx5	0/1	ImagePullBackOff	0	44s
web-nginx-748d764f7-kf9cm	0/1	ImagePullBackOff	0	46s

```
^Cjayy@DESKTOP-STVHV12:~/kubernetes$ kubectl get pods -w
```

NAME	READY	STATUS	RESTARTS	AGE
web-nginx-748d764f7-kf9cm	1/1	Running	0	111s
web-nginx-748d764f7-kwxtg	0/1	ErrImagePull	0	111s
web-nginx-748d764f7-vftx5	1/1	Running	0	111s
web-nginx-748d764f7-kwxtg	0/1	ImagePullBackOff	0	117s



Membuka Aplikasi

```
jayy@DESKTOP-STVHV12:~/kubernetes$ minikube service web-nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-nginx-service	80	http://192.168.49.2:32201

🔑 Starting tunnel for service web-nginx-service.

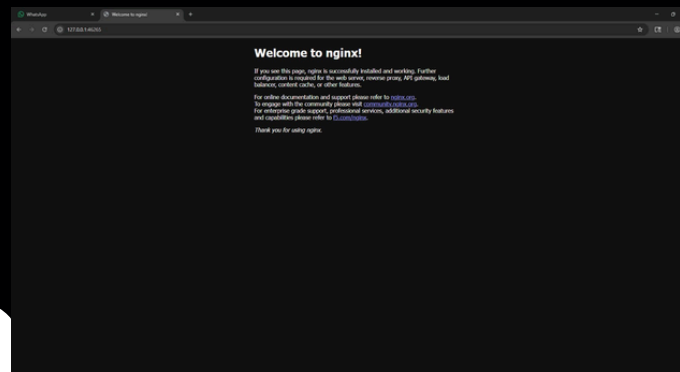
NAMESPACE	NAME	TARGET PORT	URL
default	web-nginx-service		http://127.0.0.1:43161

🌐 Opening service default/web-nginx-service in default browser...

👉 http://127.0.0.1:43161

! Because you are using a Docker driver on linux, the terminal needs to be open to run it.

^C 🛑 Stopping tunnel for service web-nginx-service.



Scaling Pod

Scaling Pod
terminal 1:

```
jayy@DESKTOP-STVHV12: ~/kubernetes$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
web-nginx-5b6cd5b8w-568cp           1/1     Running   0           4m26s
web-nginx-5b6cd5b8w-7vf3l           1/1     Running   0           4m26s
web-nginx-5b6cd5b8w-rq6ck           1/1     Running   0           4m26s
web-nginx-5b6cd5b8w-c66n6           0/1     Pending   0           0s
web-nginx-5b6cd5b8w-ht9gz           0/1     Pending   0           0s
web-nginx-5b6cd5b8w-c66n6           0/1     Pending   0           0s
web-nginx-5b6cd5b8w-ht9gz           0/1     Pending   0           0s
web-nginx-5b6cd5b8w-c66n6           0/1     ContainerCreating   0           0s
web-nginx-5b6cd5b8w-ht9gz           0/1     ContainerCreating   0           0s
web-nginx-5b6cd5b8w-ht9gz           1/1     Running    0           1s
web-nginx-5b6cd5b8w-c66n6           1/1     Running    0           1s
web-nginx-5b6cd5b8w-ht9gz           1/1     Terminating       0           33s
web-nginx-5b6cd5b8w-c66n6           1/1     Terminating       0           33s
web-nginx-5b6cd5b8w-ht9gz           1/1     Terminating       0           33s
web-nginx-5b6cd5b8w-c66n6           1/1     Terminating       0           33s
web-nginx-5b6cd5b8w-ht9gz           0/1     Completed          0           34s
web-nginx-5b6cd5b8w-c66n6           0/1     Completed          0           34s
web-nginx-5b6cd5b8w-ht9gz           0/1     Completed          0           35s
web-nginx-5b6cd5b8w-c66n6           0/1     Completed          0           35s
web-nginx-5b6cd5b8w-ht9gz           0/1     Completed          0           35s
web-nginx-5b6cd5b8w-c66n6           0/1     Completed          0           35s
web-nginx-5b6cd5b8w-ht9gz           0/1     Completed          0           35s
```

Scale Up ke 5
terminal 2:

```
jayy@DESKTOP-STVHV12: /mnt/c/WINDOWS/system32$ cd
jayy@DESKTOP-STVHV12: ~$ cd kubernetes
jayy@DESKTOP-STVHV12: ~/kubernetes$ kubectl scale deployment web-nginx --replica
s=5
deployment.apps/web-nginx scaled
jayy@DESKTOP-STVHV12: ~/kubernetes$ kubectl scale deployment web-nginx --replica
s=3
```

Lihat Deployment Live
terminal 3:

```
jayy@DESKTOP-STVHV12: ~/k$ kubectl get deployment -w
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx  3/3      3             3           5m56s
```





Menambah Worker Node



```
jayy@DESKTOP-STVHV12:~/kubernetes$ minikube node add
🐛 Adding node m02 to cluster minikube as [worker]
! Cluster was created without any CNI, adding a node to it might cause broken networking.
👉 Starting "minikube-m02" worker node in "minikube" cluster
🚢 Pulling base image v0.0.50 ...
🔥 Creating docker container (CPUs=2, Memory=2200MB) ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🌐 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔍 Verifying Kubernetes components...
🎉 Successfully added m02 to minikube!
```

```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane	16m	v1.35.1
minikube-m02	NotReady	<none>	107s	v1.35.1

```
jayy@DESKTOP-STVHV12:~/kubernetes$ minikube status
```

minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running



Menghapus Project



```
jayy@DESKTOP-STVHVI2:~/kubernetes$ kubectl delete -f deployment.yaml  
deployment.apps "web-nginx" deleted from default namespace  
jayy@DESKTOP-STVHVI2:~/kubernetes$ kubectl delete -f service.yaml  
service "web-nginx-service" deleted from default namespace  
jayy@DESKTOP-STVHVI2:~/kubernetes$ |
```





Reset Cluster Jika Error



```
jayy@DESKTOP-STVHVI2:~$ minikube start --driver=docker
🐳 minikube v1.38.1 on Ubuntu 24.04 (kvm/amd64)
🌟 Using the docker driver based on user configuration
⚠️ Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
🔥 Using Docker driver with root privileges
🚀 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.50 ...
📦 Downloading Kubernetes v1.35.1 preload ...
> preloaded-images-k8s-v18-v1...: 272.45 MiB / 272.45 MiB 100.00% 5.45 Mi
> gcr.io/k8s-minikube/kicbase...: 519.58 MiB / 519.58 MiB 100.00% 6.57 Mi
🔥 Creating docker container (CPUs=2, Memory=3072MB) ...
⚠️ Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networki
ng/proxy/
📡 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗 Configuring bridge CNI (Container Networking Interface) ...
🔍 Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
jayy@DESKTOP-STVHVI2:~$ |
```



[illegible]

```

$ kubectl describe deployment web-nginx
Name:          web-nginx
Namespace:     default
CreationTimestamp:  wed, 29 Apr 2026 08:25:54 +0700
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 1
Selector:      app=web-nginx
Replicas:      5 desired | 5 updated | 5 total | 5 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=web-nginx
  Containers:
    nginx:
      Image:      nginx:alpine
      Port:       80/TCP
      Host Port:  80/TCP
      Environment: <none>
      Mounts:      <none>
      Volumes:      <none>
      Node-Selectors: <none>
      Tolerations:  <none>
  Conditions:
    Type             Status  Reason
    ----             -
    Progressing       True    NewReplicaSetAvailable
    Available         True    MinimumReplicasAvailable
    OldReplicaSets    <none>
    NewReplicaSet     web-nginx-5b6cd3b844 (5/5 replicas created)
  Events:
    Type             Reason              Age             From             Message
    ----             -
    Normal           ScalingReplicaSet   7m40s          deployment-controller   Scaled up replica set web-nginx-5b6cd3b844 from 0 to 3

```




Describe Detail



```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl describe service web-nginx-service
Name:                web-nginx-service
Namespace:            default
Labels:               <none>
Annotations:          <none>
Selector:             app=web-nginx
Type:                 NodePort
IP Family Policy:     SingleStack
IP Families:          IPv4
IP:                   10.101.172.245
IPs:                  10.101.172.245
Port:                 <unset> 80/TCP
TargetPort:           80/TCP
NodePort:             <unset> 30519/TCP
Endpoints:            10.244.0.5:80,10.244.0.3:80,10.244.0.4:80 + 2 more...
Session Affinity:     None
External Traffic Policy: Cluster
Internal Traffic Policy: Cluster
Events:               <none>
```

```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl delete pod web-nginx-5b6cd5b844-7ppn4
pod "web-nginx-5b6cd5b844-7ppn4" deleted from default namespace
```

```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl delete pod web-nginx-5b6cd5b844-7ppn4
pod "web-nginx-5b6cd5b844-7ppn4" deleted from default namespace
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl delete deployment web-nginx
deployment.apps "web-nginx" deleted from default namespace
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl delete service web-nginx-service
service "web-nginx-service" deleted from default namespace
```

```
jayy@DESKTOP-STVHV12:~/kubernetes$ kubectl logs web-nginx-5b6cd5b844-7ppn4
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/29 01:26:38 [notice] 121: using the "epoll" event method
2026/04/29 01:26:38 [notice] 121: nginx/1.29.8
2026/04/29 01:26:38 [notice] 121: built by gcc 15.2.0 (Alpine 15.2.0)
2026/04/29 01:26:38 [notice] 121: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/04/29 01:26:38 [notice] 121: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/29 01:26:38 [notice] 121: start worker processes
2026/04/29 01:26:38 [notice] 121: start worker process 30
2026/04/29 01:26:38 [notice] 121: start worker process 31
2026/04/29 01:26:38 [notice] 121: start worker process 32
2026/04/29 01:26:38 [notice] 121: start worker process 33
2026/04/29 01:26:38 [notice] 121: start worker process 34
2026/04/29 01:26:38 [notice] 121: start worker process 35
2026/04/29 01:26:38 [notice] 121: start worker process 36
2026/04/29 01:26:38 [notice] 121: start worker process 37
2026/04/29 01:26:38 [notice] 121: start worker process 38
2026/04/29 01:26:38 [notice] 121: start worker process 39
2026/04/29 01:26:38 [notice] 121: start worker process 40
2026/04/29 01:26:38 [notice] 121: start worker process 41
2026/04/29 01:26:38 [notice] 121: start worker process 42
2026/04/29 01:26:38 [notice] 121: start worker process 43
2026/04/29 01:26:38 [notice] 121: start worker process 44
2026/04/29 01:26:38 [notice] 121: start worker process 45
2026/04/29 01:26:38 [notice] 121: start worker process 46
2026/04/29 01:26:38 [notice] 121: start worker process 47
2026/04/29 01:26:38 [notice] 121: start worker process 48
2026/04/29 01:26:38 [notice] 121: start worker process 49
2026/04/29 01:26:38 [notice] 121: start worker process 50
2026/04/29 01:26:38 [notice] 121: start worker process 51
2026/04/29 01:26:38 [notice] 121: start worker process 52
2026/04/29 01:26:38 [notice] 121: start worker process 53
```